

WEEK -4

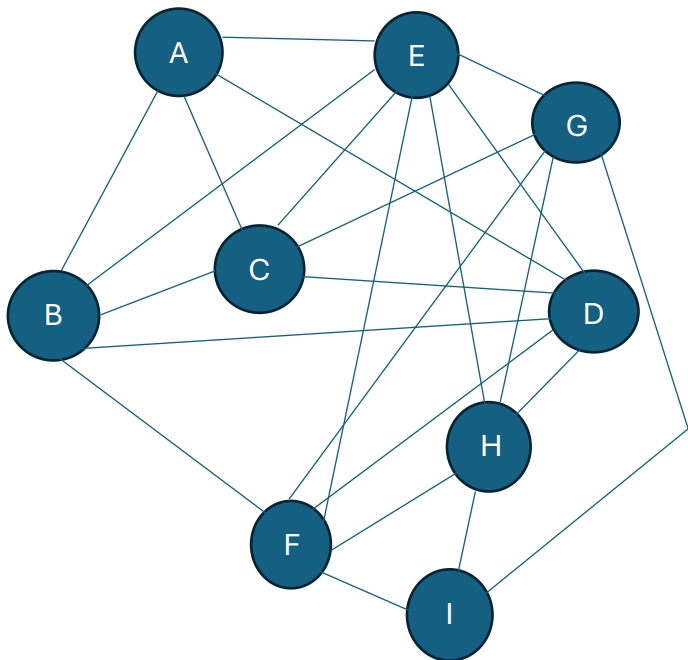
PRIMS'S & KRUSKEL'S ALGORITHM

PRIMS'S ALGORITHM:

Prim's Algorithm is used to find a Minimum Spanning Tree (MST) of a graph.

- *Connect all vertices*
- *Use no cycles*
- *Total edge cost is minimum*

GRAPH DIAGRAM:



Weight of the graph:

- i. $A-B = 4$ $A-C = 8$ $A-D = 6$ $A-E = 7$
- ii. $B-C = 5$ $B-D = 3$ $B-F = 9$ $B-E = 6$
- iii. $C-D = 4$ $C-E = 2$ $C-G = 10$
- iv. $D-E = 5$ $D-F = 7$ $D-H = 6$

v. $E-F = 4$ $E-G = 8$ $E-H = 9$
vi. $F-G = 3$ $F-H = 5$ $F-I = 6$
vii. $G-H = 4$ $G-I = 7$
viii. $H-I = 2$

CODE:

```
#include <stdio.h>
#include <limits.h>
#define V 9
int minKey(int key[], int mstSet[]) {
    int min = INT_MAX, min_index;
    for (int v = 0; v < V; v++) {
        if (!mstSet[v] && key[v] < min) {
            min = key[v];
            min_index = v;
        }
    }
    return min_index;
}
void printMST(int parent[], int graph[V][V]) {
    int cost = 0;
    printf("Edge \tWeight\n");
    for (int i = 1; i < V; i++) {
        printf("%c - %c \t%d\n", parent[i] + 'A', i + 'A',
            graph[i][parent[i]]);
        cost += graph[i][parent[i]];
    }
    printf("Total MST Cost = %d\n", cost);
}
void primMST(int graph[V][V]) {
    int parent[V];
    int key[V];
    int mstSet[V];
    for (int i = 0; i < V; i++) {
        key[i] = INT_MAX;
        mstSet[i] = 0;
    }
    key[0] = 0;
```

```

parent[0] = -1;
for (int count = 0; count < V - 1; count++) {
    int u = minKey(key, mstSet);
    mstSet[u] = 1;
    for (int v = 0; v < V; v++) {
        if (graph[u][v] && !mstSet[v] && graph[u][v] < key[v]) {
            parent[v] = u;
            key[v] = graph[u][v];
        }
    }
}
printMST(parent, graph);
}

int main() {
    int graph[V][V] = {
        {0,4,8,6,7,0,0,0,0},
        {4,0,5,3,6,9,0,0,0},
        {8,5,0,4,2,0,10,0,0},
        {6,3,4,0,5,7,0,6,0},
        {7,6,2,5,0,4,8,9,0},
        {0,9,0,7,4,0,3,5,6},
        {0,0,10,0,8,3,0,4,7},
        {0,0,0,6,9,5,4,0,2},
        {0,0,0,0,0,6,7,2,0}
    };
    primMST(graph);
    return 0;
}

```

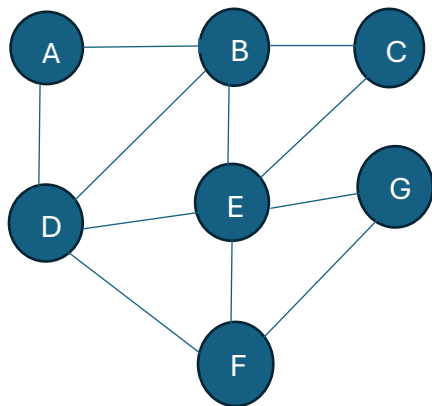
OUTPUT:

Edge	Weight
A – B	4
D – C	4
B – D	3
C – E	2
E – F	4
F – G	3
G – H	4
H – I	2
Total MST Cost = 26	

KRUSKAL'S ALGORITHM:

Kruskal's algorithm constructs a minimum spanning tree by selecting edges in ascending order of weight without forming cycles.

GRAPH DIAGRAM:



Weight of the graph:

- I. $A - B = 7$, $A - D = 5$
- II. $B - C = 8$, $B - D = 9$, $B - E = 7$
- III. $C - E = 5$
- IV. $D - E = 15$, $D - F = 6$
- V. $E - F = 8$, $E - G = 9$
- VI. $F - G = 11$

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#define V 7
#define E 11
struct Edge {
    int src, dest, weight;
};
struct Subset {
    int parent;
    int rank;
};
int find(struct Subset subsets[], int i) {
    if (subsets[i].parent != i)
        subsets[i].parent = find(subsets, subsets[i].parent);
    return subsets[i].parent;
}
void Union(struct Subset subsets[], int x, int y) {
    int xroot = find(subsets, x);
    int yroot = find(subsets, y);
    if (subsets[xroot].rank < subsets[yroot].rank)
        subsets[xroot].parent = yroot;
    else if (subsets[xroot].rank > subsets[yroot].rank)
        subsets[yroot].parent = xroot;
    else {
        subsets[yroot].parent = xroot;
        subsets[xroot].rank++;
    }
}
int compare(const void *a, const void *b) {
    return ((struct Edge *)a)->weight - ((struct Edge *)b)->weight;
}
void KruskalMST(struct Edge edges[]) {
    struct Edge result[V];
    int e = 0, i = 0;
    qsort(edges, E, sizeof(edges[0]), compare);
    struct Subset subsets[V];
    for (int v = 0; v < V; v++) {
```

```

    subsets[v].parent = v;
    subsets[v].rank = 0;
}
while (e < V - 1 && i < E) {
    struct Edge next = edges[i++];
    int x = find(subsets, next.src);
    int y = find(subsets, next.dest);
    if (x != y) {
        result[e++] = next;
        Union(subsets, x, y);
    }
}
int total = 0;
printf("Edge \tWeight\n");
for (i = 0; i < e; i++) {
    printf("%c - %c \t%d\n",
        result[i].src + 'A',
        result[i].dest + 'A',
        result[i].weight);
    total += result[i].weight;
}
printf("Total MST Cost = %d\n", total);
}

int main() {
    struct Edge edges[E] = {
        {0,1,7},
        {0,3,5},
        {1,2,8},
        {1,3,9},
        {1,4,7},
        {2,4,5},
        {3,4,15},
        {3,5,6},
        {4,5,8},
        {4,6,9},
        {5,6,11}
    };
    KruskalMST(edges);

```

```
    return 0;  
}
```

OUTPUT:

Edge	Weight
A - D	5
C - E	5
D - F	6
A - B	7
B - E	7
E - G	9
Total MST Cost = 39	